

Project Report Format

1. INTRODUCTION

1.1 Project Overview

This project shows how to import data from a spreadsheet (.xlsx/.csv) into a platform table using Import Sets and Transform Maps. The spreadsheet is first loaded into a temporary staging table (the import set table). A transform map then matches the source columns to the fields of the target table and moves the records across. Field mappings, coalesce rules, and transform scripts keep the data accurate and free of duplicates.

1.2 Purpose

- Replace slow, error-prone manual data entry with automated bulk import.
- Keep data consistent by mapping and validating fields during transformation.
- Prevent duplicate records with coalesce fields.
- Give administrators a repeatable process for future imports.

2. IDEATION PHASE

2.1 Problem Statement

Organizations often keep records in spreadsheets, and entering them one by one into a system is time-consuming and leads to typos, duplicates, and inconsistent formats.

Administrators need a reliable way to load large volumes of spreadsheet data into the correct tables with the right field mappings.

2.2 Empathy Map Canvas

Quadrant Details

Says "Entering this data manually takes forever." "Why are there duplicate records?"

Thinks "Will the columns map to the right fields?" "What if the import breaks existing data?"

Does Prepares spreadsheets, uploads files, checks records one by one, fixes errors manually

Feels Frustrated by repetitive work, anxious about data quality, relieved when the import is automated

2.3 Brainstorming

Ideas considered:

1. Manual entry through forms (rejected: slow and error-prone).
2. Custom script to insert records (rejected: needs coding for every import).
3. Import Set with Transform Map (selected): built in, configurable, reusable, and supports validation.
4. Scheduled or automated imports (kept for future scope).

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage User Action Experience / Pain

Point

Opportunity

Prepare Clean and format the
spreadsheet

Inconsistent column

names and values

Use a standard

template

Upload Load the file into the import
set table

Wrong file type or

format errors

Clear file guidelines

Map Create the transform map

and field mappings

Confusing mappings Auto-map plus
manual review

Transfor

m

Run the transform Errors or skipped rows Review the transform
history log

Verify Check the target table Missing or duplicate
data

Coalesce and

validation

3.2 Solution Requirement

Functional

- Upload an Excel/CSV file to create an import set table.
- Create a transform map linking the import set table to the target table.
- Map source fields to target fields (auto-map and manual).
- Set a coalesce field to update existing records and prevent duplicates.
- View transform results and error logs.

Non-functional

- Accurate, consistent data after import.
- Reasonable performance for large files.
- Restricted to users with admin/import roles.
- Reusable transform map for repeat imports.

3.3 Data Flow Diagram

[Spreadsheet (.xlsx/.csv)]

| upload



[Data Source]

| load data



[Import Set Table (staging)]

| transform map (field mapping, coalesce, scripts)



[Target Table]



[Transform History / Error Log]

3.4 Technology Stack

- Platform: [ServiceNow Personal Developer Instance]
- Modules: System Import Sets, Transform Maps, Data Sources

- Input format: Microsoft Excel (.xlsx) / CSV
- Target table: [e.g., Incident / User / custom table]
- Scripting (optional): JavaScript (field map and onBefore/onAfter transform scripts)

4. PROJECT DESIGN

4.1 Problem Solution Fit

The problem is bulk, error-prone manual data entry. Transform maps fit because they are built into the platform, need little or no code, and handle mapping, validation, and duplicate prevention in one process. They suit administrators who regularly move spreadsheet data into the system.

4.2 Proposed Solution

1. Prepare the spreadsheet with clear column headers.
2. Create a data source and load the file into an import set table.
3. Create a transform map from the import set table to the target table.
4. Map fields (auto-map, then adjust by hand) and set the coalesce field.
5. Run the transform and review the results.
6. Check the target table and fix any errors.

4.3 Solution Architecture

The architecture has three layers:

- Source layer: the spreadsheet file and the data source record.
- Staging layer: the import set table, which holds raw data unchanged.
- Transformation layer: the transform map (field maps, coalesce, scripts) that writes cleaned data into the target table, with the transform history recording every run.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase Activity Duration

Week

1

Requirement study, environment setup [X days]

Week

2

Spreadsheet preparation, data source and import set

creation

[X days]

Week

3

Transform map creation, field mapping, coalesce setup [X days]

Week

4

Testing, error fixing, documentation [X days]

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Test Case	Input	Expected Result	Actual Result
-----------	-------	-----------------	---------------

Result

Status

s

Upload valid

spreadsheet

.xlsx with [N] rows Import set table created [Fill in] Pass

Field mapping Auto-map columns Fields map correctly [Fill in] Pass

Transform run [N] records All rows inserted [Fill in] Pass

Duplicate

handling

Repeat file with

coalesce on

Existing records updated,

no duplicates

[Fill in] Pass

Invalid data Blank or

wrong-format

values

Rows flagged in the log [Fill in] Pass

Load time [N] records Completes in [X] seconds [Fill in] Pass

7. RESULTS

7.1 Output Screenshots

Add screenshots of:

1. The spreadsheet used as input
2. The data source and loaded import set table
3. The transform map with field mappings
4. The transform history showing inserted/updated counts
5. The target table with the imported records

8. ADVANTAGES & DISADVANTAGES

Advantages

- Saves time on bulk data entry
- Reduces human error
- Prevents duplicates through coalesce
- Reusable for future imports
- Little or no coding needed

Disadvantages

- Spreadsheet must be well-formatted and clean
- Incorrect mappings can load bad data
- Very large files may run slowly
- Needs admin-level access and some platform knowledge

9. CONCLUSION

Transform maps give a dependable way to move spreadsheet data into platform tables. By staging the data first and then mapping, validating, and coalescing it, the process cuts manual effort, improves accuracy, and keeps records consistent. The project showed that even large datasets can be imported quickly and repeatably.

10. FUTURE SCOPE

- Scheduled, automated imports from shared locations or FTP
- Advanced transform scripts for data cleansing and validation
- Import from other sources (REST APIs, databases, JDBC)
- Email notifications for import success or failure
- A dashboard tracking import history and error rates

11. APPENDIX

- Source Code: [Transform script snippets, if used]
- Dataset Link: [Link to your spreadsheet]
- GitHub & Project Demo Link: [GitHub repo URL] | [Demo video URL]